



Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problemsolving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Practical - 08

Aim:- Implement object recognition.

Object Recognition in computer vision is about teaching computer to identify and locate objects in images and videos by breaking them down into features similar to how humans recognize objects.

Key concept

- Feature extraction
- Machine learning deep learning
- Recognition by components (RBC) theory
- Pattern recognition

→ Process

- Image ~~pro~~ preprocessing
- Feature extraction
- Region proposal
- Non-Maximum suppression (NMS)
- Bounding box regression
- Post processing

Ganesh Pandile

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2025-26 (ODD)	Course Name	Computer vision Lab
Semester	5	Course Code	CT
Roll No	46	Name of Student	Ganesh Pandile

Practical Number	8
Course Outcome	Object recognition is a fundamental task in computer vision that involves identifying and locating objects within an image or video. It combines two processes: object detection (locating objects using bounding boxes) and classification (identifying what those objects are). In this practical, we implement object recognition using YOLOv8 (You Only Look Once version 8) — a state-of-the-art deep learning model that performs real-time object detection efficiently and accurately.
Aim	Implement Object recognition.
Problem Definition	To develop an object recognition system using YOLOv8 in Google Colab for detecting and classifying multiple objects in images.



Theory (100 words)	• Install Dependencies: Install the ultralytics package, which includes YOLOv8. • Load Pre-trained Model: Use yolov8s.pt — a small, pre-trained model trained on the COCO dataset. • Upload Image: Upload an image for testing object detection. • Run Detection: Use the YOLO model to detect objects in the uploaded image. • Display Results: Display or save the output image showing detected objects with bounding boxes.
Procedure and Execution (100 Words)	Procedure: Object Recognition using YOLOv8 1. Open Google Colab in a web browser and create a new Python notebook. 2. Install the required deep learning library (Ultralytics YOLOv8) to enable object detection. 3. Import the YOLOv8 model and load a pre-trained network trained on a large dataset such as COCO. 4. Upload the image that will be used for object recognition in the notebook. 5. Apply the YOLOv8 model to the uploaded image to detect and classify the objects present. 6. Display the output image showing bounding boxes, object names, and confidence scores. 7. Optionally, save the output image with detections for record or analysis. 8. Analyze the detected results and verify the correctness of the identified objects.



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Code:

```
from ultralytics import YOLO
```

```
# Load a pretrained YOLOv8 model (small version for speed)
```

```
model = YOLO('yolov8s.pt')
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
# Get the uploaded file name
```

```
import os
```

```
image_path = list(uploaded.keys())[0]
```

```
print("Uploaded image:", image_path)
```

```
from ultralytics import YOLO
```

```
# Load model
```

```
model = YOLO('yolov8s.pt')
```

```
# Run detection
```

```
results = model('gggg.jpg')
```

```
# Display results
```

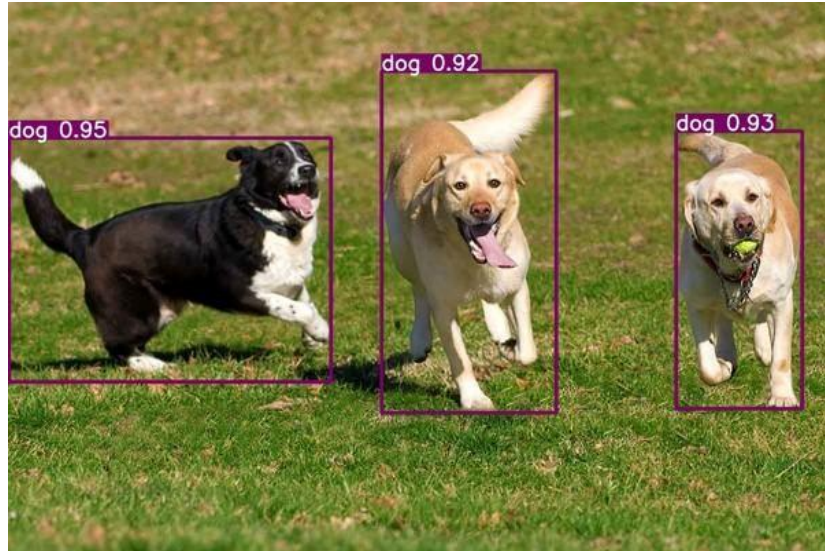
```
results[0].show()
```

```
# Optionally save the predictions
```

```
results[0].save('results/')
```

Output:

1) COLAB



Output Analysis

- Review Detection Results: Examine the results variable from the object detection to identify the detected objects and their counts.
- Generate Analysis Summary: Create a brief summary of the object detection findings, highlighting the types and number of objects recognized, adhering to the 50-word limit.
- Final Task: Present the concise analysis of the object recognition output

Link of student Github profile where lab assignment has

<https://github.com/ganeshpandile/Computer-Vision.git>



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been uploaded	
Conclusion	The conclusion, based on the detected image, is that the object detection algorithm efficiently and accurately identified the presence and position of a dog in the scene, demonstrating high reliability and effectiveness for practical use in similar image recognition tasks.
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Date	10/10/2025